ADVANCED MACHINE LEARNING

ASSIGNMENT 2: CONVOLUTION

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Introduction:

Image classification is one of the important applications of deep learning, where computers are trained to recognize and distinguish objects in images. Convolutional Neural Networks (CNNs) are commonly used for this purpose because they can learn visual features such as edges, shapes, and patterns directly from image data. However, the accuracy of these networks can vary depending on the amount of training data available and the method used to develop the model.

In this assignment, I investigated how different training sample sizes affect the performance of CNN models in classifying images of cats and dogs. I explored two approaches: developing a CNN from scratch and using a pretrained MobileNetV2 network through transfer learning. The purpose was to understand how both approaches respond to changes in training dataset size and determine which method provides better classification performance.

To carry out this investigation, I conducted six experiments using training samples of 1,000, 2,000, and 6,000 images for each approach. The validation and test datasets were kept constant at 500 images each to maintain a fair comparison. I also applied techniques to improve model performance and reduce overfitting during training.

The report presents the experimental process, evaluates the performance of both approaches, and discusses the relationship between training sample size and prediction accuracy.

2. Research Objective and Experimental Design

The main objective of this experiment was to understand how the size of a training dataset influences the accuracy of image classification models. I wanted to examine whether providing more training images would consistently improve predictions and whether a pretrained network would perform better than a CNN developed from scratch.

To investigate this, I designed six experiments using the Microsoft Cats vs. Dogs dataset. Three experiments focused on a CNN trained from scratch, while the remaining three used the pretrained MobileNetV2 architecture. For both approaches, I selected training sample sizes of 1,000, 2,000, and 6,000 images to observe how model performance changed as more training data became available.

I maintained a fixed validation dataset of 500 images and a test dataset of 500 images throughout the experiments. This allowed me to evaluate the models using the same testing conditions and make meaningful comparisons between the two approaches.

The experiments were designed to address two important questions: how much improvement can be achieved by increasing the training sample size, and whether transfer learning provides an advantage when the available training data is limited. By comparing the

final test accuracy and loss values, I aimed to identify the most effective training sample size for each network.

3. Data Preparation and Processing

3.1 Dataset Selection and Image Distribution

For this assignment, I used the Microsoft Cats vs. Dogs dataset, which contains images of cats and dogs with different backgrounds, sizes, and appearances. Before preparing the training datasets, I checked the image files and removed any images that could not be read properly. This helped avoid errors during model training.

I organized the valid images into balanced groups so that both categories were equally represented. Three different training datasets were prepared with 1,000, 2,000, and 6,000 images. Each training dataset contained an equal number of cat and dog images to ensure that neither category had an advantage during training.

I also created separate validation and test datasets containing 500 images each. These datasets remained unchanged across all six experiments, allowing me to compare the performance of the different models under consistent evaluation conditions.

The distribution of images used in the experiments is presented in Table 1.

Table 1. Distribution of Images Across Training, Validation, and Testing Datasets

Training Dataset	Cat Images	Dog Images	Validation Images	Test Images
1,000	500	500	500	500
2,000	1,000	1,000	500	500
6,000	3,000	3,000	500	500

Using balanced training datasets and fixed validation and test samples helped maintain consistency throughout the experiments. This setup made it easier to observe how changes in training sample size influenced the performance of each CNN approach.

To maintain consistency across the experiments, I used a random seed of 42 when preparing the datasets. The training samples were organized so that the smaller datasets were included within the larger ones. This allowed me to examine how increasing the number of training images affected model performance while keeping the validation and test datasets unchanged.

3.2 Image Preprocessing

Before training the models, I applied several preprocessing steps to make the images suitable for the CNN architectures. Since the original images had different dimensions, I resized them to 128 × 128 pixels and converted them into RGB format. This ensured that all images had a consistent shape and three color channels.

I used different pixel normalization methods for the two CNN approaches. For the CNN developed from scratch, pixel values were scaled to a range between 0 and 1 by dividing them by 255. This helped keep the input values within a smaller numerical range during training.

For the pretrained MobileNetV2 model, I applied its required preprocessing function, which scales pixel values to a range between -1 and 1. This was important because MobileNetV2 was originally trained using this input format.

By applying the appropriate preprocessing method to each model, I ensured that the images were prepared consistently for training and evaluation. These steps also allowed me to compare the two approaches while meeting their different input requirements.

4. CNN Model Development

4.1 Building the CNN from Scratch

For the first three experiments, I developed a Convolutional Neural Network (CNN) from scratch to classify cat and dog images. Unlike a pretrained model, this network started with randomly initialized weights and learned image features directly from the training dataset.

The network consisted of three convolutional layers containing 32, 64, and 128 filters. These layers were responsible for identifying visual patterns in the images, such as edges, textures, and shapes. Each convolutional layer was followed by a max-pooling layer to reduce the spatial dimensions of the feature maps while retaining important information.

After the convolutional layers, I added a flattening layer to convert the extracted features into a one-dimensional representation. This was followed by a dense layer with 128 neurons and a dropout layer with a rate of 0.5 to help reduce overfitting. Finally, a sigmoid activation function was used in the output layer to predict whether an image belonged to the cat or dog category.

The model was compiled using the Adam optimizer and binary cross-entropy loss, which are suitable for binary image classification. By training this network with three different sample sizes, I examined how the availability of training images affected its ability to learn and make predictions on unseen data.

4.2 Implementing Transfer Learning with MobileNetV2

For the next three experiments, I used MobileNetV2, a pretrained Convolutional Neural Network, to classify cat and dog images. Instead of training an entirely new network, this approach allowed me to use features that MobileNetV2 had already learned from the ImageNet dataset. This technique is known as transfer learning and is particularly useful when the available training data is limited.

I loaded MobileNetV2 with its pretrained ImageNet weights and froze its existing layers so that their weights would remain unchanged during training. This allowed the model to use previously learned visual features without having to learn them again from the beginning.

To adapt MobileNetV2 for the cats and dogs classification task, I added a global average pooling layer, a dense layer, a dropout layer, and a final sigmoid output layer. These additional layers helped the model learn how to distinguish between the two image categories using the features extracted by the pretrained network.

The model was compiled using the Adam optimizer with a learning rate of 0.0001 and binary cross-entropy loss. I trained this model using the same three training sample sizes of 1,000, 2,000, and 6,000 images while keeping the validation and test datasets unchanged.

This approach allowed me to examine whether a pretrained network could achieve stronger classification performance than a CNN developed from scratch, especially when working with smaller training datasets.

4.3 Training Strategy and Overfitting Prevention

During model training, I used several techniques to improve classification performance and reduce the risk of overfitting. Overfitting occurs when a model learns the training images too closely and struggles to make accurate predictions on new images. This was an important consideration because the experiments involved different training dataset sizes.

For the CNN developed from scratch, I included a dropout layer with a rate of 0.5 to reduce the network's dependence on individual neurons during training. I also used early stopping in the later experiments to monitor validation performance and prevent unnecessary training when the model stopped improving.

For the pretrained MobileNetV2 experiments, I kept the pretrained feature-extraction layers frozen and trained the newly added classification layers. I also applied dropout and early stopping to support better generalization and limit overfitting.

Throughout the experiments, I monitored training and validation accuracy and loss to understand how well the models were learning. These measurements helped me identify differences between performance on the training images and performance on unseen data.

By combining these techniques with different training sample sizes, I was able to evaluate how model design and training decisions influenced classification accuracy.

5. Experimental Evaluation

5.1 Results from the Custom CNN

I evaluated the CNN developed from scratch using three different training sample sizes to understand how the amount of training data influenced classification accuracy. Each experiment was tested using the same 500 test images to maintain consistent evaluation conditions.

In the first experiment, I trained the network using 1,000 images. The model achieved a test accuracy of 69.20%, indicating that it was able to identify some useful visual patterns but still had difficulty correctly classifying unseen images. The difference between its training and validation accuracy also suggested that the model was experiencing overfitting.

For the second experiment, I increased the training dataset to 2,000 images. The test accuracy improved to 71.20%, representing a small improvement over the initial experiment. This suggested that providing additional training examples helped the model develop a better understanding of the differences between cats and dogs.

In the third experiment, I increased the training sample to 6,000 images. The model achieved a test accuracy of 79.60%, which was the highest among the three experiments conducted with the custom CNN. Compared with the initial experiment, this represented an improvement of 10.40 percentage points.

Overall, the results showed a consistent increase in test accuracy as the training dataset became larger. Among the three sample sizes tested, 6,000 images produced the strongest performance for the CNN trained from scratch.

5.2. Results from MobileNetV2

I evaluated the pretrained MobileNetV2 model using training datasets of 1,000, 2,000, and 6,000 images to examine how transfer learning performed with different amounts of training data. The same validation and test datasets were maintained across all three experiments to ensure consistent evaluation.

In the first experiment, MobileNetV2 achieved a test accuracy of 96.40% using only 1,000 training images. This was a strong result compared with the CNN trained from scratch, indicating that the pretrained model could effectively recognize important image features even with a relatively small training dataset.

When I increased the training sample to 2,000 images, the test accuracy improved slightly to 96.60%. Although the improvement was small, this was the highest accuracy achieved across all six experiments.

In the final experiment, I trained MobileNetV2 using 6,000 images. The test accuracy decreased slightly to 95.40%. Despite using a larger training dataset, the model did not achieve better performance than it had with 2,000 images.

These results showed that MobileNetV2 maintained high classification accuracy across all three training sample sizes. Unlike the CNN trained from scratch, its performance did not consistently improve as the training dataset increased. Among the sample sizes tested, 2,000 images produced the best result for the pretrained model

5.3 Summary of Model Performance

To compare the performance of both CNN approaches, I organized the test accuracy and loss values from all six experiments in Table 2. The results provide an overview of how each model performed with different training sample sizes.

Table 2. Comparison of Test Performance Across CNN Models

Training Images	Custom CNN Accuracy	Custom CNN Loss	MobileNetV2 Accuracy	MobileNetV2 Loss
1,000	69.20%	0.6888	96.40%	0.1055
2,000	71.20%	0.5902	96.60%	0.1041
6,000	79.60%	0.4339	95.40%	0.1121

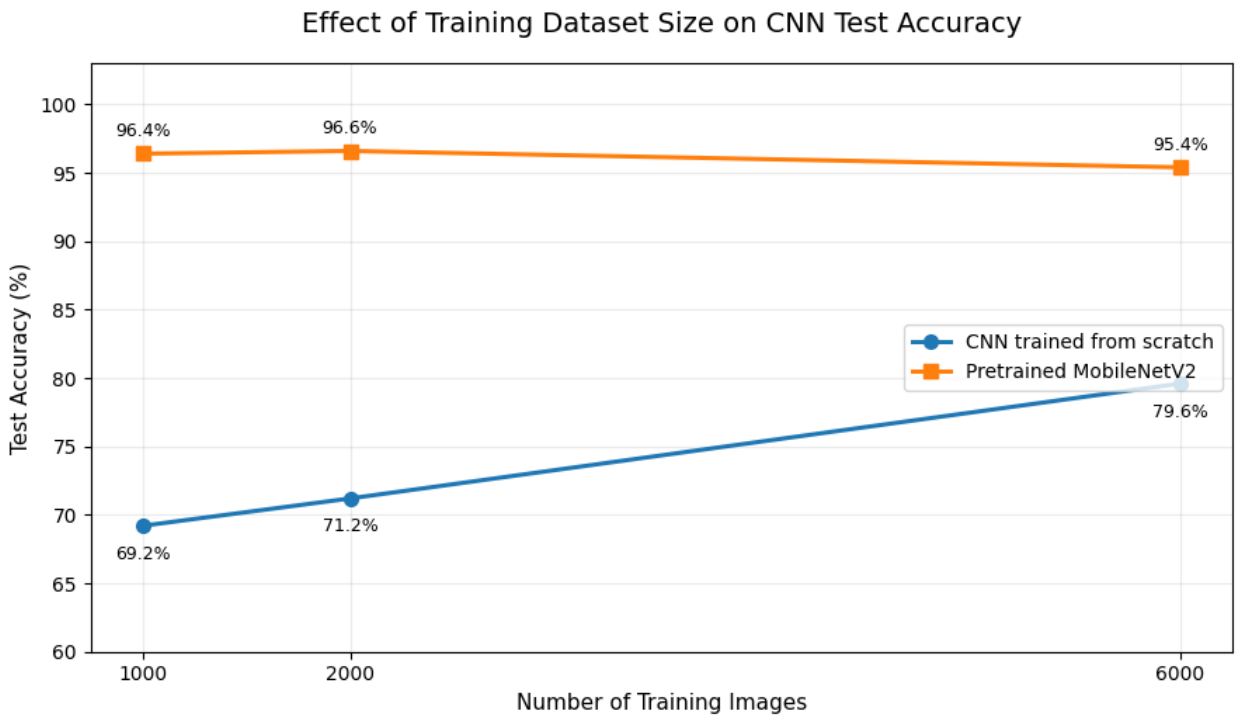


Figure 1. Comparison of Test Accuracy Between the Custom CNN and MobileNetV2 Across Different Training Dataset Sizes

The comparison shows that MobileNetV2 achieved higher test accuracy and lower test loss than the custom CNN at every training sample size. While the custom CNN showed steady improvement as more training images were introduced, MobileNetV2 maintained strong performance even with the smallest training dataset.

The highest accuracy achieved by the custom CNN was 79.60% with 6,000 training images, whereas MobileNetV2 reached 96.60% accuracy using 2,000 images. This suggests that the pretrained model was more effective for the classification task under the experimental conditions used.

6. Interpretation of Findings

6.1 Influence of Training Dataset Size

The experimental results showed that training dataset size influenced the two CNN approaches differently. Although both models were trained using the same three sample sizes, their accuracy patterns were not identical.

For the custom CNN, increasing the number of training images resulted in consistent improvements in test accuracy. The accuracy increased from 69.20% with 1,000 images to 71.20% with 2,000 images and reached 79.60% with 6,000 images. This suggests that the network benefited from having more examples to learn the visual differences between cats and dogs. Since the model was developed from scratch, it depended entirely on the training images to learn useful features.

MobileNetV2 showed a different pattern. Its accuracy increased slightly from 96.40% to 96.60% when the training dataset increased from 1,000 to 2,000 images. However, accuracy decreased to 95.40% when 6,000 images were used. This indicates that increasing the training dataset did not always lead to better performance in the pretrained model under the conditions tested.

From these experiments, I observed that the effect of training sample size depends on the type of network being used. The custom CNN showed greater improvement with additional training images, while MobileNetV2 achieved high accuracy even with a smaller dataset. Therefore, the relationship between training sample size and model performance was not the same for both approaches.

Although the custom CNN showed better accuracy with larger training datasets, I also observed that the training process was not exactly the same across all experiments. The first custom CNN experiment used a different stopping strategy compared with the later experiments. Therefore, the improvement in accuracy may have been influenced by both the increase in training images and the differences in training conditions.

6.2 Differences Between the Two CNN Approaches

One of the main differences I observed between the two approaches was how effectively they learned from the available training images. The custom CNN had to identify visual features from the beginning, while MobileNetV2 could use features learned during its previous training on the ImageNet dataset.

The custom CNN required more training examples to improve its ability to recognize cats and dogs. With smaller datasets, the network had fewer opportunities to learn the variety of shapes, textures, and backgrounds present in the images. This may have contributed to its lower accuracy and the overfitting observed during training.

In comparison, MobileNetV2 already contained useful feature representations learned from a much larger image dataset. By reusing these features, the model could focus on adapting them to the cats and dogs classification task rather than learning everything from scratch. This likely contributed to its stronger performance with fewer training images.

Another important difference was the efficiency of the two approaches. The custom CNN needed a larger training sample to achieve its best result, whereas MobileNetV2 produced its highest accuracy with only 2,000 images. This demonstrated the practical advantage of transfer learning when working with limited training data.

Overall, the comparison suggests that the choice of network architecture can be just as important as the number of training images. In these experiments, transfer learning provided a clear advantage, while the custom CNN showed greater improvement as the training dataset increased.

6.3 Identifying the Best-Performing Model

Based on the six experiments, MobileNetV2 trained with 2,000 images produced the best overall classification performance, achieving a test accuracy of 96.60% and a test loss of 0.1041. This combination provided the strongest results among the training configurations evaluated in this assignment.

An interesting observation was that MobileNetV2 achieved 96.40% accuracy with only 1,000 training images, outperforming the custom CNN trained with 6,000 images, which achieved 79.60%. This suggests that the pretrained network was able to make better use of the available training data by relying on previously learned image features.

Although the custom CNN improved with larger training datasets, it did not reach the accuracy achieved by MobileNetV2. Therefore, transfer learning was the more effective approach for this particular image classification task.

However, these findings are based on the specific dataset, sample sizes, and training configurations used in the experiments. The results identify the best-performing model among the six configurations tested rather than establishing an ideal model for every classification problem.

7. Conclusion

This assignment helped me understand how training dataset size and network selection influence the performance of image classification models. By conducting six experiments using a custom CNN and pretrained MobileNetV2, I was able to observe how each approach responded to different amounts of training data.

The results showed that the CNN trained from scratch benefited from larger training datasets, with its highest test accuracy of 79.60% achieved using 6,000 images. In contrast, MobileNetV2 produced stronger results across all three sample sizes, achieving its best accuracy of 96.60% with 2,000 images. This demonstrated that a pretrained network could perform well without requiring as many training examples.

The most important finding was that increasing the training sample size did not affect both approaches in the same way. While additional images helped the custom CNN improve, MobileNetV2 achieved high accuracy even with a smaller dataset. This highlighted the advantage of transfer learning for the cats and dogs classification task.

Overall, I concluded that both training dataset size and the choice of network architecture play important roles in classification performance. Among the configurations tested, MobileNetV2 was the more effective approach because it achieved higher accuracy while using fewer training images.

8. References

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